

CO2: MCQ Test — Answer Key

Course: Artificial Intelligence (CSA17) | Assessment Tool 3

Q.No	Correct Answer	Reasoning
1	C) Greedy Best-First Search	Uses only $h(n)$, ignoring cost so far.
2	C) $f(n) = g(n) + h(n)$	A*'s evaluation function.
3	B) Admissibility	Heuristic never overestimates true cost.
4	B) Hill Climbing	A local search algorithm.
5	B) Minimax Algorithm	Used for adversarial decision-making.
6	B) Greedy Best-First Search	Decides based only on estimated distance.
7	B) A* Search	Combines cost so far + estimated remaining cost.
8	B) Constraint Satisfaction Problem	Variables, domains, constraints (exam scheduling).
9	C) Minimax Algorithm	Evaluates moves and anticipates opponent responses.
10	B) Online Search Agent	Decides step-by-step in an unknown environment.
11	B) It ensures optimality	Admissible heuristic guarantees A* finds the optimal solution.
12	B) Reduces number of nodes evaluated	Alpha-Beta pruning's main benefit.
13	C) Getting stuck in local maxima	Main drawback of Hill Climbing.
14	C) Constraint satisfaction	A valid CSP solution satisfies all constraints.
15	B) Estimate utility of a state	Role of an evaluation function.
16	A) 12	$f(n) = g(n) + h(n) = 7 + 5 = 12$.
17	B) A* becomes non-optimal	Overestimating heuristic breaks admissibility.
18	C) 31	Full binary tree depth 4: $2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 1 + 2 + 4 + 8 + 16 = 31$.
19	B) 27	Leaf nodes = branching factor ^{depth} = $3^3 = 27$.
20	B) $O(b^{(d/2)})$	Best-case time complexity of Alpha-Beta pruning.